

Embedded AI

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Foundations and Basic Tools

Overall Roadmap of the Course and Developing the Right Mindset

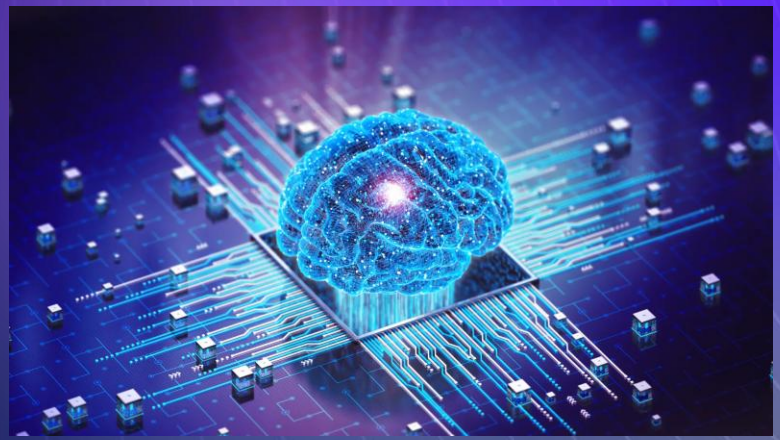
Introduction

"We are about to embark on a journey where the power of Artificial Intelligence is brought into the compact, fast, and energy-efficient world of microcontrollers. Our goal is to deploy intelligent models with our own hands on real hardware."



What is Embedded AI?

The integration of Machine Learning (ML) models with embedded systems (microcontrollers) and bringing intelligent capabilities to small and resource-constrained hardware



Why Embedded AI?

Latency: On-device decision-making ensures nearly zero delay.
Privacy: Data never leaves the device.
Energy Efficiency: Leveraging low-power processors significantly extends battery life

Where is it Deployed?

Cloud AI: High computational power, but high latency and internet dependency
Edge AI: Closer to the data source, offering faster response
In-Browser AI: Model execution directly within the browser for quick interaction

